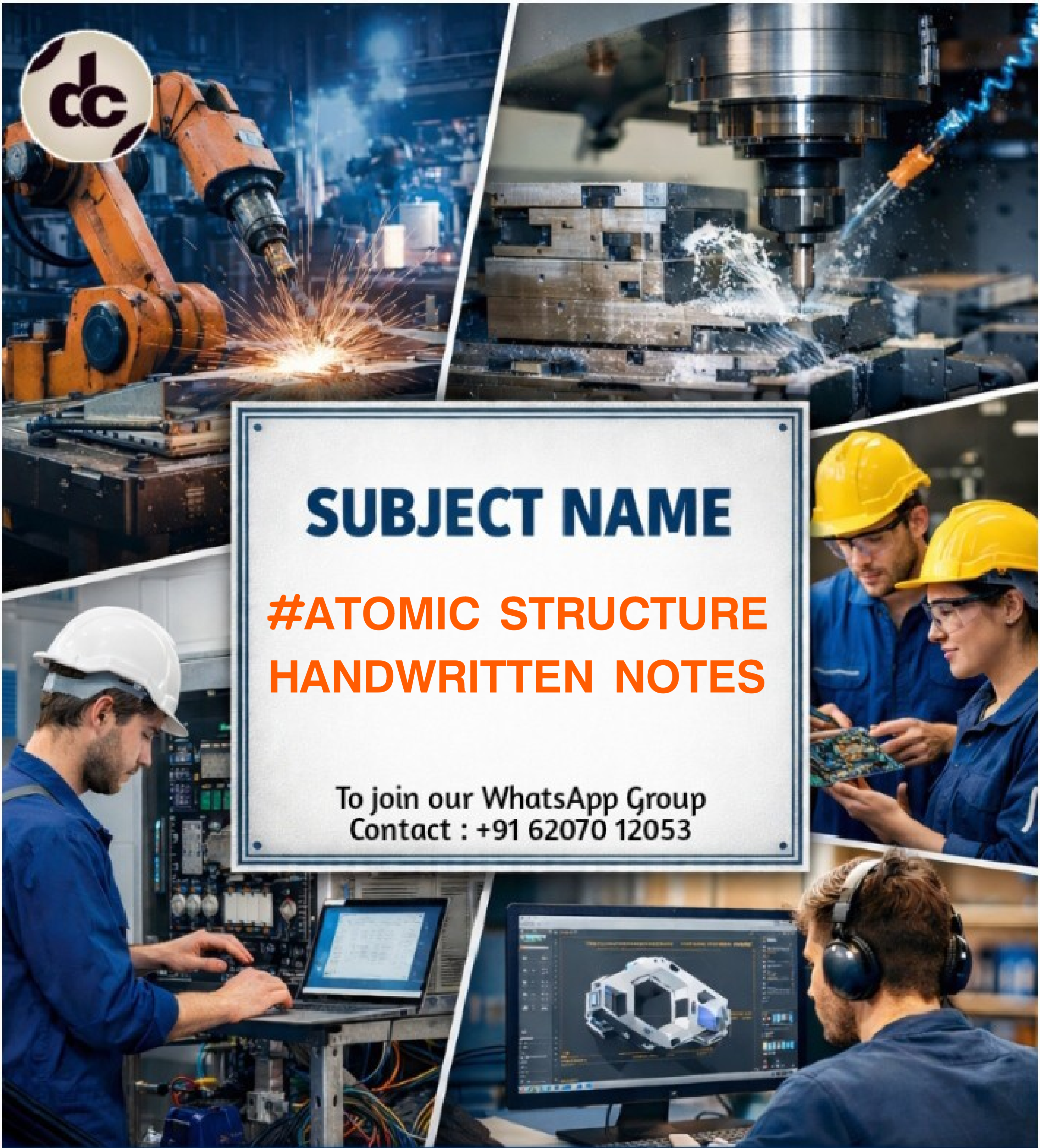

SUBJECT NAME

#ATOMIC STRUCTURE HANDWRITTEN NOTES

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DIPLOMA CAMPUS

Unit - I Atomic Structure -

→ Fundamental Particles:

An atom consists of three fundamental particles, electron, proton & neutron.

Fundamental particles	Symbol	Mass	Charge	Discover
electron	${}^{0}_{-1}e$ or e^-	9.109×10^{-31} kg	1.602×10^{-19} C	J.J Thompson
proton	${}^{1}_{+1}P$ or ${}^{1}_{1}H$	1.672×10^{-27} kg	$+1.602 \times 10^{-19}$ C	Rutherford
neutron	${}^{1}_{0}n$	1.6748×10^{-27} kg	No charge	J. Chadwick

1.602×10^{-19} Coulomb = unit Charge.

$+1.602 \times 10^{-19}$ Coulomb is considered to be +1 called unit ~~charge~~ positive charge. -1.602×10^{-19} C is considered to be −1, called unit negative charge.

→ Rutherford model of Atom (Rutherford's α-Particle scattering experiment)

A narrow beam of α-particles having high energy from a radioactive source was directed at thin foil of gold.

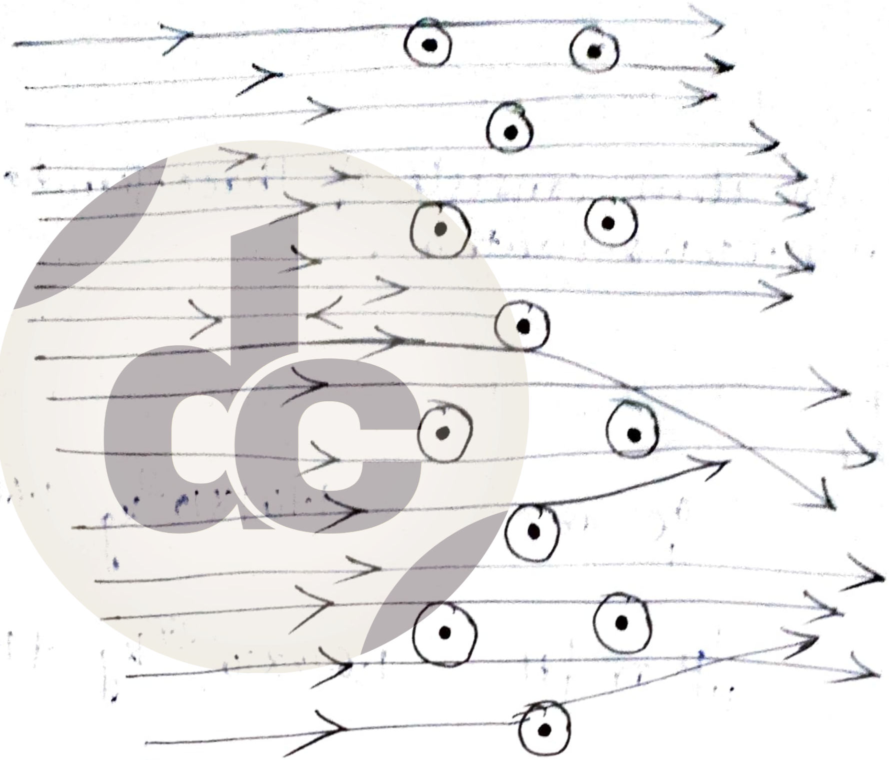
Gold foil surrounded by circular flourescent zinc sulphide screen. When α-particles hit the screen, tiny flash of light was produced.

Observation:

i) Most of the α-particles passed through the gold foil without suffering any deflection from straight path.

ii) Small fraction of the α-particles was deflected by small angles.

iii) very few, ~1 in 20,000 bounced back, were deflected by 180°.



Conclusion:

i) Most of the space in the atom is empty, as most of the α-particles passed through the foil undeflected.

ii) Few +vely charged particles were deflected, due to repulsive force of the positive charge of the atom, that is concentrated in a very small volume.

iii) α-particles which are likely to hit the nucleus, rebound back and only a small no. of α-particles rebound as the size of nucleus is very small.

Proposal of nuclear model of atom-

i) The positive charge and most of the mass of the atom is concentrated in extremely small region. This small region is called nucleus.

ii) The nucleus is surrounded by e^-, which move around the nucleus ~~at~~ very high speed ~~&~~ in circular paths called orbits.

iii) Electrons (e^-) & nucleus are held together by electrostatic force of attraction.

Drawbacks of Rutherford model-

i) Inability to explain the stability of atom:

According to Maxwell's electromagnetic theory, an accelerating charged particle (electron) will continuously emit energy in the form of radiation i.e. kinetic energy of the electron will gradually go on decreasing. As a result the balance between the centrifugal force and Coulombic force of attraction will be disturbed and the electron will fall upon the nucleus following a spiral path. Thus, there will be no existence of e^- in an atom.

ii) Inability to explain the line spectra of the elements.

iii) Inability to describe ~~discribe~~ distribution of e^- & energies of e^-.

Bohr's Atomic Model-

To overcome the limitations of Rutherford's atomic model, Neil Bohr proposed his atomic model with the help of planck's quantum theory.

Postulates-

i) Electrons revolves only in those orbits which have a fixed value of energy. Hence, these orbits are called energy levels or stationary states. As long as an electron revolves in these circular paths (also called orbits) no energy is emitted or absorbed.

ii) Atom consists of a small, heavy positively charged species called nucleus, in the centre and e^- revolve around it in circular orbits.

iii) when an electron jumps from one energy level to the other, energy is either emitted or absorbed.

iv) Absorption or emission of energy occurs in the form of quanta ($h\nu$)

v) The permissible circular orbits are those which satisfy the equation -

$$mvr = n\left(\frac{h}{2\pi}\right)$$, where h = Planck's constant.

mvr = angular momentum

n = principal quantum no.

Quantization of energy-

when a quantity cannot change gradually and continuously to have any arbitary value but changes only abruptly and discontinuously to have certain definite or discrete value, are called as quantized.

Usefulness -

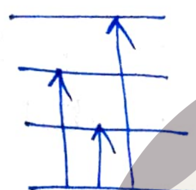
i) It explains the stability of atom. -

· e^- cannot lose energy as long as it stays in a particular orbit. Therefore, the question of losing energy continuously and falling into the ~~orbits~~ nucleus doesnot arise.

ii) It explains the line spectrum of hydrogen and H-like particles.

e^- neither emits nor absorbs energy as long as it stays in a particular orbit.

When subjected to electric discharge or high temperature



e^- moves from g.s to higher energy level i.e excited state.

· Lifetime is short, e^- will travel back to lower energy levels or g.s.

During each jump, energy is emitted in the form of a photon of light of a definite wavelength or frequency.

$E_2 - E_1 = h\nu$, $\Delta E = h\nu = h\frac{c}{\lambda}$

'h' is Planck's constant.

Appearance of lines in the hydrogen spectrum-

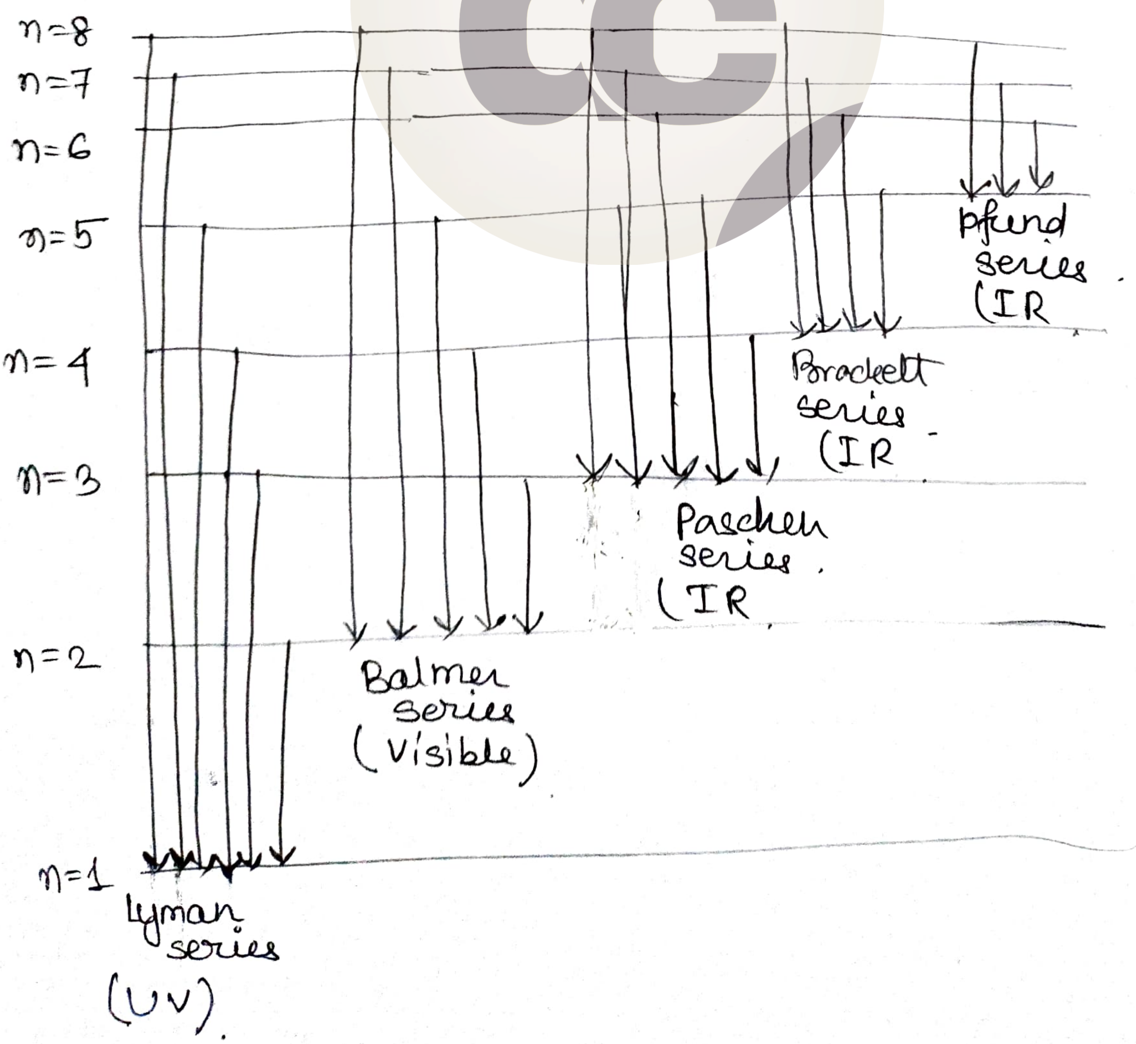
Lyman, Balmer, Paschen, Brackett, Pfund.

Any given sample of Hydrogen gas contains large number of molecules. When subjected to heat energy or electric discharge the molecules split into atoms. e⁻ in different H- atom absorb different amt of energy and are excited to different energy levels.

Some are excited to 2nd energy level (L)
" " " " 3rd " " (M).
" " " " 4th " " (N).

Life time in higher energy level is small, so the e⁻ return back to lower energy level or ground level in one or more jumps.

Various possibility-



different excited e^- adopt different routes to return to va... lower energy levels or ground state. Emits different amou... of energies and thus produce large no. of lines in the atomic Spectrum of H-

Limitations of Bohr's Model of Atom -

i) Inability to explain line spectra of multi electron atoms.

Explained line Spectra for H- and H-like atom containing Single e^-, ~~but~~ but failed to explain for multi-e^- system.

With high resolution ~~micr~~ spectroscopes, it was found that each line was split up into a no. of closely spaced lines, fine structures, could not be explained by Bohr's model.

ii) Inability to explain Splitting of lines in the magnetic field (Zeeman effect) and in the electric field (Stark effect).

If the source emitting the radiation is placed in a magnetic field or electric field, each line is observed ~~to~~ to split into no. of lines. Bohr could not explain this.

iii) Inability to explain the 3D model of Atom.

~~At~~ Bohr proposed that e^- move along certain circular paths in one plane. This gives a flat model of atom.

iv) Inability to explain the shapes of molecules.

v) Inability to explain de-Broglie's concept of dual nature of matter ~~&~~ Heisenberg's uncertainty principle.

Heisenberg's Uncertainty Principle

Accurate measurement of position & Velocity for sub-atomic particles is not possible.

As a consequence of dual nature of matter, Heisenberg proposed a principle & it states that:

It is impossible to measure simultaneously the position and momentum of small particle with absolute accuracy or certainty. If an attempt is made to measure any of these two quantities with higher accuracy, the other becomes less accurate. The product of the uncertainty in the momentum and uncertainty in position is always constant and is equal to or greater than $h/4\pi$, where 'h' is Planck's constant, ie.,

$$\Delta x \cdot \Delta p \geq h/4\pi \quad \text{---(i)}$$

$$\Rightarrow \Delta x \cdot \Delta p \simeq \frac{h}{4\pi} \quad \text{---(ii)}$$

Momentum, is given by,

$$\Delta p = m \times \Delta v$$

Putting the value of Δp in equation (i),

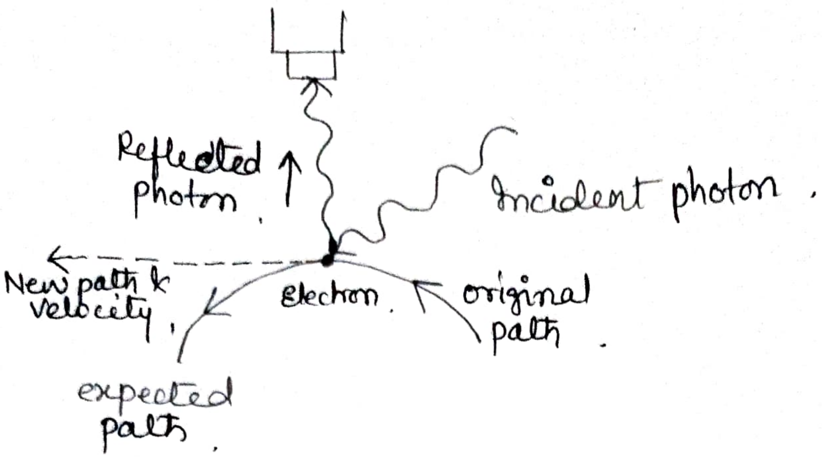
$$\Delta x \cdot (m \Delta v) \geq \frac{h}{4\pi}$$

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

This implies that position and velocity of a particle cannot be measured simultaneously with certainty.

$\Rightarrow$ uncertainty principle applies to position and momentum along the same axis.

Explanation-



i) To measure the position & momentum of an electron, the e^- is hit with light (photon).

ii) photon strike the e^- and reflected photon is seen in the microscope.

iii) due to hitting position as well as velocity of e^- changes.

iv). Accuracy of the position depends on the ~~ω~~ λ of light used. uncertainty in position is $\pm\lambda$.

v) Shorter λ, greater position accuracy.

vi) Shorter λ, higher frequency & hence higher energy. High energy photon will change the speed as well as direction of e^-.

vii) greater λ, ~~small~~ greater accuracy in speed, but lower accuracy in position.

Significance of Heisenberg's uncertainty principle-

Significant only for sub atomic particle.

Energy of photon is insufficient to change the position & velocity for bigger bodies. For ex- light to football.

For ex- any matter of 1mg.

$$\Delta x . \Delta v = \frac{h}{4\pi m}.$$

$$= \frac{6.626 \times 10^{-34}\ kg\, m^2 s^{-1}}{4 \times 3.1416 \times 10^{-6}\ kg}.$$

$\approx 10^{-28}\ m^2 s^{-1}$, value is very small hence negligible.

For ex- e^-

$$\Delta x . \Delta v = \frac{h}{4\pi m} = \frac{6.626 \times 10^{-34}\ kg\, m^2/s}{4 \times 3.1416 \times (9.11 \times 10^{-31}\ kg)}$$

$\approx 10^{-3}\ m^2/s$, significant value.

why e^- cannot reside in the nucleus?

Diameter of nucleus $= 10^{-15} m$.

If e^- reside inside the nucleus, the maximum uncertainty in its position would have been $10^{-15} m$

$$\Delta x = 10^{-15} m.$$

mass of $e^- = 9.1 \times 10^{-31}\ kg$.

minimum uncertainty in velocity is.

$$\Delta x . \Delta p = \frac{h}{4\pi}.$$

$$\Delta x (m \Delta v) = \frac{h}{4\pi}.$$

$$\Delta v . \Delta x = \frac{h}{4\pi m} = \Delta v = \frac{h}{4\pi m . \Delta x} = \frac{6.6 \times 10^{-34}\ kg\, m^2/s}{4 \times 3.1416 \times (10^{-15} m) \times (9.1 \times 10^{-31}}$$

$= 5.77 \times 10^{10}\ m/s$.

antum Numbers -

The characteristics, like shape, size and orientation (direction), are ~~des~~ expressed in terms of three number principal, azimuthal & magnetic quantum no.

1. Principal Quantum no. -

The most important quantum no., since it tells the principal energy level or shell to which the electron belong.

'n' ⇒ denoted by.

Can have any integral value except zero,

$n = 1, 2, 3, 4 \ldots$

designated by K, L, M, N .....

This number helps to explain the main lines of the spectrum on the basis of the electronic jumps between these shells.

Information:

i) gives the average distance of the e^- from the nucleus. (size of e^- cloud)

ii) determines the energy of the e^- in H-atom & H-like particles (which contain only one e^-).

$$E_n = -\frac{2\pi^2 m e^4}{n^2 h^2}$$

m = mass
e = charge of e^-
h = planck's constant

For multielectron atoms, it determines only the approximate energy of any e^-.

Energy order.

K < L < M < N < O < P.

1 < 2 < 3 < 4 < 5 .....

iii) Maximum no. of e^- present in any principal shell is given by $2n^2$ where n is the no. of the principal shell.

2. Azimuthal Quantum No:

This number helps to explain the fine lines of the spectrum.

Info:

i) no. of subshells present in the main shell.

ii) Angular momentum of the e^- present in any subshell.

iii) the relative energies of the various subshells.

iv) Shapes of the various subshells present within the same principal shell.

denoted by letter 'l'.

given value of n, it can have any integral value ranging from 0 to n-1,

1st shell (K), n=1

l=0 ⇒ s-subshell.

These notations are the initial letters of the words sharp, principa[l], diffused and fundamental

2nd shell (L), n=2

l=0 & 1 ⇒ p - subshell.

3rd shell (M), n=3

l = 0, 1, 2 ⇒ diffused

4th shell (N), n=4

l= 0, 1, 2, 3 ⇒ fundamental.

Energy order,

$s<p<d<f$.

...gnetic Quantum No.

Explains Zeeman Effect.

orbital motion around nucleus generates electric field. This in turn produces a magnetic field which can interact with the external magnetic field.

Under the influence of external magnetic field, ~~which can~~ ~~under~~ the e^- of a subshell can orient themselves in certain preferred regions of space, called orbitals.

Determines no. of orbitals presents in any subshell,

denoted by m,

m = +l to -l including zero,

thus for every value of l, m has 2l+1.

for, l = 0, s-subshell, m = 1, s-orbital.

l = 1, p-subshell, m = +1, 0, -1, p_x-orbital, p_y, p_z } d.o

l = 2, d-subshell, m = +2, +1, 0, -1, -2, $\underbrace{d_x, d_y, d_z, d_{xy}, d_{x^2-y^2}}_{\text{degenerate orbital (same energy)}}$

4. spin quantum no: denoted by m_s.

In an orbital, e^- can either ~~m~~ spin in clockwise or anticlockwise direction, hence m_s can have two values, $+\frac{1}{2}$ or $-\frac{1}{2}$, or represented by ↑ or ↓.

Pauli Exclusion Principle-

No two e^- in an atom can have the same set of four quantum nos.

This principle is called exclusion principle.

Thus, it follows from the above discussion that in an atom, any two electrons may have the same values for any of the three quantum nos. but the fourth must be different.

$n=3, l=0, m=0, m_s=+1/2$ or $-1/2$.

$n=3, l=0, m=0, m_s=+1/2$

$n=3, l=0, m=0, m_s=-1/2$

Thus, it is given as

An orbital can have a maximum two e^- and these must have opposite spin.

No. of subshells in nth shell $= n$

No. of orbitals in a subshell $= 2l+1$

Max. no. of e^- in a subshell $= 2(2l+1)$

No. of orbitals in nth shell $= n^2$

Max. no. of e^- in nth shell $= 2n^2$

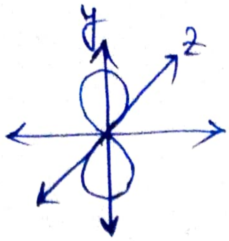
Paschen, Brackett and Pfund.

Shell	Principal (n)	l	m	m_s	Electrons present	Total No. of e^- present
K	1	0	0	$+\tfrac12, -\tfrac12$	2	2
L	2	0 (2s)	0	$+\tfrac12, -\tfrac12$	2	8
		1 (2p)	−1	$+\tfrac12, -\tfrac12$	6	
			0	$+\tfrac12, -\tfrac12$		
			+1	$+\tfrac12, -\tfrac12$		
M	3	0 (3s)	0	$+\tfrac12, -\tfrac12$	2	18
		1 (3p)	−1, 0, +1	$+\tfrac12, -\tfrac12$; $+\tfrac12, -\tfrac12$; $+\tfrac12, -\tfrac12$	6	
		2 (3d)	−2, −1, 0, +1, +2	$+\tfrac12, -\tfrac12$ ... $+\tfrac12, -\tfrac12$	10	
N	4	0 (4s)	0	$+\tfrac12, -\tfrac12$	2	32
		1 (4p)	−1, 0, +1	$+\tfrac12, -\tfrac12$; $+\tfrac12, -\tfrac12$; $+\tfrac12, -\tfrac12$	6	
		2 (4d)	−2, −1, 0, +1, +2	$5(+\tfrac12, -\tfrac12)$	10	
		3 (4f)	−3, −2, −1, 0, +1, +2, +3	$7(+\tfrac12, -\tfrac12)$	14	

Shapes of Atomic Orbitals -

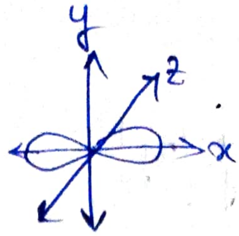
s-orbital

spherical in shape.

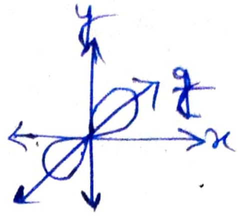


p_y

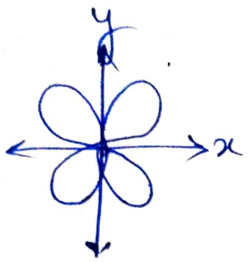
Dumb-bell shaped or lobe-shaped.



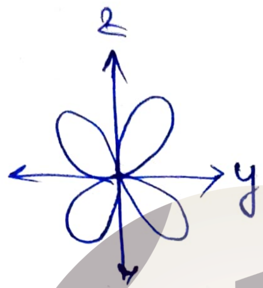
p_x



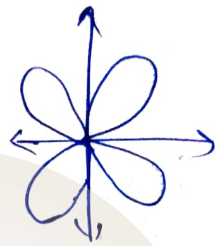
p_z



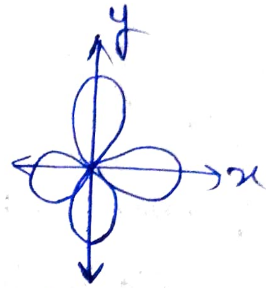
d_{xy}.



d_{yz}.

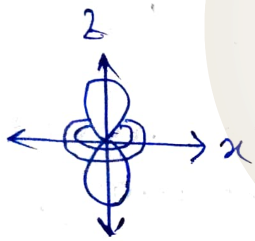


d_{xz}.



$d_{x^2-y^2}$.

clover leaf shaped.



d_{z^2}.

doughnut-shaped electron cloud in the centre.

No. of nodes in any orbital $= (n-l-1)$

Filling of orbitals -

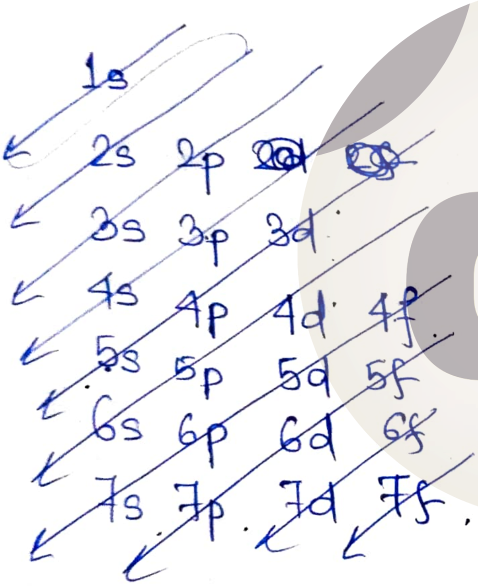
1) Aufbau Principle :-

German word ⇒ building up.

It states that -

In the ground state of the atoms, the orbitals are filled in order of their increasing energies. In other words, e^- first occupy the lowest energy orbital available to them, & enter into higher energy orbitals only when the lower energy orbitals are filled.

order.



The order is -

1s, 2s, 2p, 3s, 3p, ~~3d~~, 4s, ~~4~~ 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p....

Alternatively can be done by (n+l) rule :

In neutral isolated atom, the lower the value of (n+l) for an orbital, lower is its energy.

If two different types of orbitals have the same value of (n+l), the orbital with lower value of 'n' has lower energy.

For ex-

Type of orbital	Value of n	Value of l	Value of $n+l$	Relative Energy
1s	1	0	1	Lowest
2s	2	0	2	higher than 1s
2p	2	1	3	$n=2$, thus lower energy than 3s, where $n=3$.
3s	3	0	3	
3p	3	1	4	3p has lower energy than 4s, $n=3$ & $n=4$
4s	4	0	4	

An orbital can have maximum two electrons and these must have opposite spins.

Correct: $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\downarrow\uparrow}$ $\boxed{\downarrow\uparrow}$ one clockwise revolving other anticlockwise.

Incorrect: $\boxed{\uparrow\uparrow}$ $\boxed{\downarrow\downarrow}$ In this case all the quantum numbers are similar, which is violating Pauli's exclusion principle.

Hund's Rule of Maximum Multiplicity:

Electron pairing in p, d, & f orbitals cannot occur until each orbital of a given subshell contains one electron each or is singly occupied.

Electrons repel each other when present in the same orbital.

Minimised by 1st filling the orbitals single.

Either clockwise ~~an~~ or anti-clockwise.

For ex-

1s	2s	$2p_x$	$2p_y$	$2p_z$
↑↓	↑↓	↑	↑	↑

Total no. of paired $e^- = 4$ (2 pairs)

Total no. of unpaired $e^- = 3$.

Electronic Configuration of Elements -

The distribution of electrons into different shells, Subshells & orbitals of an atom is called its electronic configuration.

orbitals represented by box.

Hydrogen - At. No. $= Z = 1$.

One e-

$1s^1$ or 1s ↑

Helium $(Z=2)$. At. No. $= Z = 2$

2e-

$1s^2$ $1s^2$ ↑↓

$1s^2$ → No. of electron; 1 → principal quantum no.; s → Subshell.

Lithium $(Z=3)$, $e^- = 3$

$1s^2 2s^1$ $1s^2$ ↑↓ $2s^1$ ↑

Beryllium $Z=4$, $e^- = 4$

$1s^2$ ↑↓ $2s^2$ ↑↓

Shell or orbit:

The circular paths surrounding the nucleus along which e^- revolve without emission of energy are called shells or orbits. They are designated as K-shell, M-shell, N shell etc.

Atomic number & Mass number:

Atomic number is the number of protons within the nucleus of an atom. It is also equal to the no. of e^- in the atom.

No. of protons in nucleus = No. of e^- in the neutral atom.

To maintain electrical neutrality, the no. of protons & no. of e^- should always match in a neutral atom.

~~Mass~~, No. of protons in H = 1, So Atomic number = 1.
" " " in Na = 11, So Atomic no. = 11.

Atomic Number is denoted by 'Z'.

Mass no. is the total no. of protons & neutrons present within the nucleus of an atom.

C^{12} has 6 protons & 6 neutrons, thus, mass no. of Carbon = 6+6 = 12.

Isobar -

Atoms of different elements having same mass number are called isobar.

For ex - $^{40}_{18}Ar$, $^{40}_{19}K$, $^{40}_{20}Ca$

$^{14}_{6}C$, $^{14}_{7}N$.

Isotopes - atoms with identical atomic no. but different mass no. are called isotopes.

~~For ex~~

For ex – Protium (^1_1H) $\longrightarrow$ 99.985%

Deuterium (^2_1D) $\longrightarrow$ 0.015%

Tritium (^3_1T) $\longrightarrow$ Trace

$\underbrace{^{12}_{6}C,\ ^{13}_{6}C,\ ^{14}_{6}C}_{\text{Isotopes}}$, $\underbrace{^{35}_{17}Cl,\ ^{37}_{17}Cl}_{\text{Isotopes}}$

Isotones –

Different atoms having same no. of neutrons are called isotones.

For ex – $^{30}_{14}Si$, $^{31}_{15}P$

No. of neutrons for $^{30}_{14}Si$ = (30 − 14) = 16

No. of neutrons for $^{31}_{15}P$ = (31 − 15) = 16